

A Federated Autoencoder Framework With Explainable AI for Intelligent 6G-IoT Infrastructure Optimization

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Abstract—Sixth generation (6G) wireless networks with ultralow latency, high reliability, and massive connectivity require intelligent and privacy-concerned infrastructure optimization. This work presents a federated autoencoder platform combined with explainable artificial intelligence (XAI) for performance optimization of 6G-Internet of Things (IoT) systems. The method integrates traditional machine learning algorithms (decision tree, random forest (RF), logistic regression, AdaBoost, gradient boosting (GB)) with a variational autoencoder (VAE) for dimensionality reduction and feature extraction. Federated learning (FL) is utilized to maintain data privacy among distributed edge nodes, and SHAP and LIME explainers are utilized for explaining model decisions at the local and global levels. The framework points out key Quality-of-Service (QoS) parameters like latency and throughput as major optimization levers. Experimental outcomes on the 6G-IoT dataset indicate that RF with highest accuracy for 80:20 split and GB has a 99.8% accuracy in a 10-fold validation, and FL gets a ROC-area under curve (AUC) value of 0.999 with robust privacy guarantees. XAI enhances

transparency and regulatory compliance by making attribution of predictions to contributing features. As a whole, the proposed approach provides an interpretable, privacy-conscientious, and scalable tool for intelligent 6G-IoT infrastructure management.

Index Terms—Artificial intelligence (AI), explainable AI (XAI), federated learning (FL), Internet of Things (IoT) networks, sixth generation (6G) optimization, variable auto encoders (VAE).

I. INTRODUCTION

AS 5G matures and researchers look forward to sixth generation (6G) wireless systems for unprecedented capability (terabit-per-second data rates, ultralow latency, extreme connectivity), the Internet of Things (IoT) is bursting to billions of devices (sensors, cars, wearables) gathering data [1], [2]. To meet 6G-specific goals like URLLC, mMTC, and eMBB outlined in IMT-2030, the proposed framework targets latency, throughput, and connectivity optimization—core enablers of 6G readiness [3], [4]. In this work, there is focus on optimizing key network parameters—minimizing latency and packet loss, while maximizing throughput and energy efficiency—to enhance 6G-IoT infrastructure performance.

Current 6G-IoT optimization techniques often rely on either centralized learning or static rule-based systems, which are not adaptive to dynamic network conditions or heterogeneous device behaviors. 6G vision statements (e.g., IMT-2030) and artificial intelligence (AI) regulation (e.g., the EU AI Act) enunciate that imperative for responsible AI [5], [6]. Federated learning (FL) is a privacy strategy well-suited to 6G-IoT's decentralized topology [7]. Rather than centralized training (risk of data leaks), FL stores data locally and transmits model updates. Sensitive data remains local, significantly mitigating privacy threats and network overhead. However, FL alone does not guarantee interpretability of the learned models, which is essential for regulatory compliance, especially under policy frameworks like GDPR and the EU AI Act. To bridge this gap, the proposed work integrates explainable AI (XAI) methods that provide transparency across distributed nodes, enabling explainable decision support at both global and local levels. To provide explainability and accountability, XAI techniques incorporated for human-interpretable model explanations [8]. This is in line with upcoming 6G regulatory frameworks, enabling standards like ITU-T FG-AI4H and IEEE P7001 with model-agnostic tools, such as SHAP and LIME [9]. In contrast

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to previous work restricted to post-training analysis or partial understanding, the proposed end-to-end system incorporates local FL training, instance- and global-level explanations, and practical Quality-of-Service (QoS) optimization [10].

The main contributions of this study are outlined as follows.

- 1) A privacy-preserving 6G-IoT optimization framework is proposed by integrating FL with XAI.
- 2) A comparative evaluation is conducted on ML and FL models. (random forest (RF) and gradient boosting (GB) with highest accuracy in various folds and FL with highest ROC).
- 3) The proposed system identifies key optimization parameters latency and throughput which are critical for enhancing network performance.

II. RELATED WORK

AI is at the center of the new 6G-IoT paradigm, enabling intelligent network management and autonomic IoT services with extreme connectivity and latency support [11], [12], [13]. Two of the leading research areas are FL for decentralized, private model learning and XAI for model transparency. Yet there is still only a single trustworthy AI framework for 6G-IoT in its infancy [14], [15].

FL facilitates edge device training by transmitting updates instead of raw data, which suits privacy requirements in industrial IoT, healthcare, and smart homes [16], [17], [18]. This has fueled research on client selection, model compression, and communication-efficient optimization to scale and robustify FL [19].

At the same time, XAI has come to the forefront in interpreting closed-box models within IoT systems. Models, such as LIME and SHAP provide explanations for single predictions based on local behavior approximation or attribution of importance of features and are widely applied in anomaly detection and network analysis [20], [21], [22], [23], [24].

IoT analytics has shifted from traditional models (i.e., decision trees (DTs), SVMs) toward deep learning models, such as CNNs, RNNs, and variational autoencoders (VAEs) [25], [26], [27], [28].

In spite of individual advancements, FL and XAI are seldom together. FL studies mostly ignore interpretability, whereas XAI takes centralized data for granted [29].

Most FL-XAI studies rely on centralized post-hoc explanations and do not extend well to federated settings. The proposed work is a complete FL-XAI pipeline with SHAP and LIME for both global and local interpretability, alongside a VAE-based encoder and multiobjective optimization approach not yet fully explored in 6G-IoT research.

III. RESEARCH GAP

FL research focuses on accuracy and privacy but lacks interpretability, while XAI tools like SHAP and LIME rarely address decentralized, non-IID 6G-IoT settings. Moreover, there is not much work that brings together classical and deep models—specifically, VAEs in a federated and explainable framework using actual 6G-IoT data. The proposed work is a end-to-end framework that combines FL, VAE, and XAI into

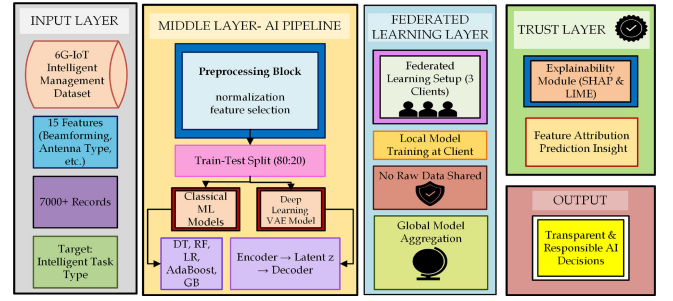


Fig. 1. Overall pipeline combining preprocessing, classical models, VAE, FL, and SHAP/LIME-based explainability.

a privacy-conscious and interpretable optimization pipeline customized to 6G QoS goals.

IV. METHODOLOGY

This section presents the complete architecture and mathematical formulation of the proposed trustworthy AI system for 6G-IoT. The proposed model integrates classical and deep learning models with FL and XAI, as visualized in Figs. 1–3. The key components include preprocessing, model training, privacy-preserving FL, SHAP/LIME explainability, and final decision synthesis.

A. System Architecture Overview

Fig. 1 illustrates the end-to-end framework. The system inputs the 6G-IoT Intelligent Management Dataset [30] (15 features, 7000+ samples), does preprocessing, splits the data, and trains classical models (DT, RF, logistic regression (LR), AdaBoost, GB) as well as a VAE. Training is split into three clients via FL. Three clients were selected to represent computational constraints and edge heterogeneity, modeling small-scale deployments common in nascent 6G testbeds. Missing values were not present class imbalance managed through stratified splitting for fair distribution during training.

B. Preprocessing and Feature Normalization

Each feature vector $\mathbf{x}^{(j)} \in \mathbb{R}^d$ is normalized

$$x_{\text{norm}}^{(j)} = \frac{x^{(j)} - \mu^{(j)}}{\sigma^{(j)}} \quad (1)$$

where $\mu^{(j)}$ and $\sigma^{(j)}$ denote the mean and standard deviation (SD) of the characteristic j th throughout the data set. Categorical features are one-hot encoded. Oversampling or undersampling methods, such as SMOTE were not used since the dataset was fairly balanced between classes.

C. Classical Machine Learning Models

The classical interpretable models are used to benchmark QoS optimization in 6G-IoT. These models were selected for their low complexity, ease of integration with XAI, and stable performance in resource-constrained environments. RF was specifically chosen for its compatibility with TreeSHAP and strong validation stability, making it ideal for feature-level explainability.

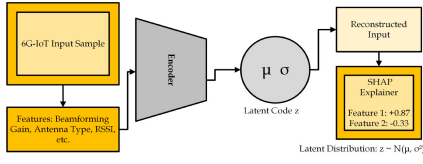


Fig. 2. VAE encoder-decoder pipeline with SHAP-based explainability on the reconstruction.

- 1) *DT*: Splits data based on Gini impurity

$$G(S) = 1 - \sum_{c=1}^C p_c^2. \quad (2)$$

- 2) *RF*: An ensemble of T DTs

$$\hat{y}_{RF} = \text{mode}(f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_T(\mathbf{x})). \quad (3)$$

- 3) *LR*: Models the class probability

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}}}. \quad (4)$$

- 4) *AdaBoost*: Updates weights based on misclassification

$$\alpha_t = \frac{1}{2} \ln\left(\frac{1 - \epsilon_t}{\epsilon_t}\right). \quad (5)$$

- 5) *GB*: Minimizes loss via gradient descent

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta h_m(\mathbf{x}). \quad (6)$$

D. Variational Autoencoder

The VAE is used to learn compressed representations of IoT features (Fig. 2). Input $\mathbf{x}$ is mapped to latent space $\mathbf{z}$ via encoder

$$\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (7)$$

where

$$\boldsymbol{\mu} = f_{\text{enc}}^{\boldsymbol{\mu}}(\mathbf{x}), \quad \boldsymbol{\Sigma} = \exp(f_{\text{enc}}^{\boldsymbol{\Sigma}}(\mathbf{x})). \quad (8)$$

Reconstruction loss

$$\mathcal{L}_{\text{rec}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2. \quad (9)$$

KL-divergence

$$\mathcal{L}_{\text{KL}} = \frac{1}{2} \sum_{k=1}^d (\mu_k^2 + \sigma_k^2 - \log \sigma_k^2 - 1). \quad (10)$$

Total VAE loss

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{rec}} + \beta \mathcal{L}_{\text{KL}}. \quad (11)$$

VAE compresses high-dimensional IoT features into low-dimensional latent vectors suited for federated training, offering noise reduction and generalization with fewer parameters than CNNs/LSTMs. The loss weight β was fixed at 0.1 to balance reconstruction and regularization, ensuring lightweight, privacy-preserving learning.

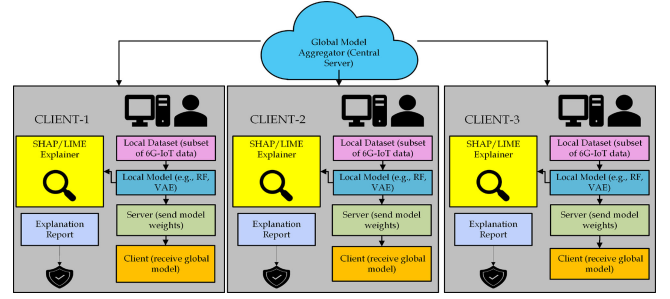


Fig. 3. Federated training with SHAP/LIME explainability at each client node.

E. Federated Learning Architecture

As shown in Fig. 3, FL is deployed across 3 clients. Let θ_i^r be the model parameters on client i at round r . Local update is

$$\theta_i^{r+1} = \theta_i^r - \lambda \nabla_{\theta} \mathcal{L}_i(\theta_i^r). \quad (12)$$

In (12), λ is the learning rate, $\mathcal{L}_i$ is the local loss, n_i is the number of samples on client i , and N is the total across all clients.

Global aggregation via FedAvg

$$\theta^{r+1} = \sum_{i=1}^K \frac{n_i}{N} \theta_i^{r+1}. \quad (13)$$

Communication rounds continue until convergence

$$\|\theta^{r+1} - \theta^r\|_2 < \epsilon. \quad (14)$$

F. Explainable AI: SHAP and LIME

SHAP assigns each feature a contribution value ϕ_k

$$\phi_k = \sum_{S \subseteq F \setminus \{k\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{k\}) - f(S)] \quad (15)$$

where F is the set of all features.

LIME fits a linear surrogate model $g(\mathbf{x})$

$$g(\mathbf{x}) = \boldsymbol{\alpha}^T \mathbf{x} + \gamma. \quad (16)$$

The local fidelity loss minimized is

$$\mathcal{L}_{\text{LIME}} = \sum_j \pi_j (\hat{f}(\mathbf{x}_j) - g(\mathbf{x}_j))^2 + \Omega(g) \quad (17)$$

where π_j is a locality kernel and $\Omega(g)$ is model complexity.

SHAP and LIME were applied post-hoc at each client to explain local model outputs—TreeSHAP for tree models and KernelSHAP for VAE. LIME used a cosine kernel with 500 perturbations. Key features like latency and throughput consistently emerged as dominant, guiding fair attribution and optimization across training rounds.

Algorithm 1 Explainable FL With XAI (LIME/SHAP) for 6G-IoT Optimization

Require: K clients, initial global model w_0 , total rounds T

- 1: **for** each round $t = 1$ to T **do**
- 2: Server sends current model w_{t-1} to selected clients
- 3: **for** each client $k \in \mathcal{S}_t$ in parallel **do**
- 4: Receive w_{t-1} from server
- 5: Train model locally using local data $\mathcal{D}_k$
- 6: Compute local model w_k^t by minimizing:

$$\mathcal{L}_k(w) = \frac{1}{|\mathcal{D}_k|} \sum_{(x_i, y_i) \in \mathcal{D}_k} \ell(f_w(x_i), y_i)$$

- 7: Generate local explanations $\phi_k(x)$ using LIME or SHAP:

$$f(x_i) = \phi_0 + \sum_{j=1}^M \phi_j(x_i)$$

- 8: Send $(w_k^t, \phi_k(x))$ to server (or)

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]$$

- 9: **end for**
- 10: Server aggregates model updates:

$$w_t \leftarrow \sum_{k=1}^K \frac{n_k}{n} w_k^t$$

- 11: Optionally, aggregate explanations:

$$\Phi_t(x) = \frac{1}{K} \sum_{k=1}^K \phi_k(x)$$

- 12: **end for**
 - 13: **Output:** Final global model w_T and explanation profile $\Phi_T(x)$
-

G. Global Loss Function and Optimization

Across M models and K clients

$$\mathcal{L}_{\text{global}} = \frac{1}{K} \sum_{i=1}^K \left[\frac{1}{M} \sum_{m=1}^M \mathcal{L}_{i,m} \right]. \quad (18)$$

Each $\mathcal{L}_{i,m}$ corresponds to the local loss from model m on client i . Optimization continues until

$$\text{Accuracy}_{\text{global}} > \tau \quad \text{or} \quad r > r_{\text{max}}. \quad (19)$$

In (19), τ is the target global accuracy threshold and r_{max} is the maximum number of federated rounds. Weights α_1 to α_4 in the multiobjective loss were empirically optimized with grid search to achieve the proper balance between weights of latency, throughput, energy efficiency, and packet loss objectives, and the best-performing configuration was used within final training.

Algorithm IV-G represents a VAE framework that can be applied for dimensionality reduction, data uniformity encoding and data translation and regeneration is preferred.

Algorithm 2 VAE for 6G-IoT Optimization

Require: Input data x , encoder network $q_\phi(z|x)$, decoder network $p_\theta(x|z)$, learning rate η , number of epochs E

- 1: **for** epoch = 1 to E **do**
- 2: **for** each data point x_i in dataset **do**
- 3: Encode input x_i to get parameters of latent distribution:

$$(\mu, \log \sigma^2) = f_\phi(x_i)$$

- 4: Sample latent variable z using reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

- 5: Decode latent variable:

$$\hat{x}_i = g_\theta(z)$$

- 6: Compute the reconstruction loss:

$$\mathcal{L}_{\text{rec}} = \|x_i - \hat{x}_i\|^2$$

- 7: Compute the KL divergence:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^d \left(1 + \log(\sigma_j^2) - \mu_j^2 - \sigma_j^2 \right)$$

- 8: Compute total loss:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \beta \cdot \mathcal{L}_{\text{KL}}$$

- 9: Update parameters ϕ, θ using gradient descent:

$$(\phi, \theta) \leftarrow (\phi, \theta) - \eta \cdot \nabla_{\phi, \theta} \mathcal{L}$$

- 10: **end for**

- 11: **end for**

- 12: **Output:**

Trained encoder: $q_\phi(z|x)$

Trained decode: $p_\theta(x|z)$

H. Parameter-Specific Optimization Strategy

In 6G-IoT systems, performance is optimized by manipulating key network parameters. SHAP and LIME identify the most influential QoS features, guiding optimization in both centralized FL as per the Algorithm 1. Each parameter's role and objective are defined, as shown in Fig. 4, where latency and packet loss are minimized, and throughput and energy efficiency are maximized, with interpretable feedback from SHAP and LIME.

- 1) *Latency (ms)*: Represents the time delay in end-to-end data transmission. As high latency degrades user experience and real-time performance, the optimization goal is

$$\min \mathbb{E}_{i=1}^N [\text{latency}_i]. \quad (20)$$

SHAP analysis confirms latency as the most influential negative feature affecting throughput.

- 2) *Throughput (Mb/s)*: Measures the rate of successful data delivery. Higher throughput corresponds to better

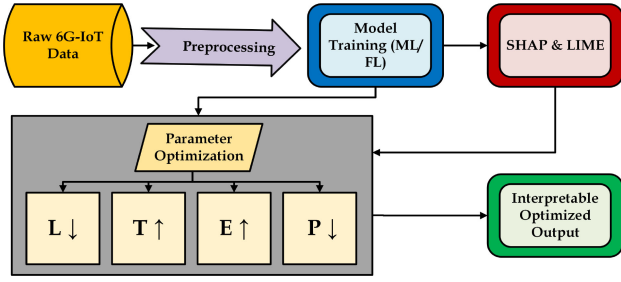


Fig. 4. System-level optimization pipeline targeting key QoS parameters in 6G-IoT.

network utility as

$$\max \mathbb{E}_{i=1}^N [\text{throughput}_i]. \quad (21)$$

- 3) *Energy Efficiency*: Especially relevant in IoT devices operating under power constraints. Defined as throughput per unit energy

$$\max \mathbb{E}_{i=1}^N \left[\frac{\text{throughput}_i}{\text{Energy Consumption}_i} \right]. \quad (22)$$

Models are trained to learn resource-efficient data delivery.

- 4) *Packet Loss Rate (%)*: High loss indicates instability and reduced QoS. Thus, minimizing this is essential

$$\min \mathbb{E}_{i=1}^N [\text{PacketLossRate}_i]. \quad (23)$$

This helps maintain data integrity and application reliability.

- 5) *Bandwidth Utilization (%)*: Refers to the percentage of total bandwidth used. The proposed framework encourages efficient use without causing overload, optimizing routing and load balancing indirectly.
- 6) *Network Slice Failure / Overload Status*: These binary indicators signify degraded network service. Models learn to minimize such occurrences by learning optimal configurations.

The individual parameter objectives are unified into a multiobjective loss function

$$\min_{\theta} \mathcal{L}_{\text{total}} = \alpha_1 \cdot L + \alpha_2 \cdot P - \alpha_3 \cdot T - \alpha_4 \cdot E. \quad (24)$$

In (24), L , P , T , and E denote latency, packet loss, throughput, and energy efficiency. Coefficients α_1 to α_4 were optimized using grid search, with SHAP and LIME providing direction to highlight important features, such as latency and throughput, to align optimization with actual real-world QoS requirements.

Here, α_1 , α_2 , α_3 , and α_4 are tunable weights reflecting the importance of each parameter. The optimizer updates the model to minimize $\mathcal{L}_{\text{total}}$ during training iterations. However, the proposed work focuses on latency and throughput as they are the two QoS parameters available in the dataset.

V. RESULTS

The results section presents the experimental setup, evaluation metrics, and analysis. Autoencoder models were implemented using Keras with TensorFlow, while other

TABLE I
PAIRED *t*-TEST RESULTS FOR THROUGHPUT AND LATENCY

Comparison	<i>t</i> statistic	<i>p</i> value
Throughput (Mbps) vs. Latency (ms)	64.878	< 0.0001

machine learning models used the Scikit-learn library. The evaluation focuses as per the available dataset, on key QoS-related features of the 6G-IoT data set, such as latency and throughput, and assesses model performance using precision, precision, recall, F1 score, and ROC-area under curve (AUC) to understand their impact on system optimization. The Paired T-test for the parameters throughput and the latency is presented in Table I. This means the difference between throughput and latency values (in raw numeric scale) is statistically significant.

The accuracy is represented as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (25)$$

The precision of the model is represented as

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (26)$$

The recall of the framework is represented as

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (27)$$

F1-score of the framework is represented as

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}. \quad (28)$$

The ROC value of the models is represented as

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR}). \quad (29)$$

ROC curves and AUC are used as key measures to quantify model discrimination and performance estimation, especially in imbalanced class settings. A high AUC value of approximately 1.0 indicates satisfactory class separation, confirming the accuracy of both centralized and federated classifiers. The ML models, such as DT, RF, LR, AB, GB are applied for the performance comparison analysis. The values are tabulated in Table II.

The RF and GB models have shown stable performance on all the metrics and DT, AB also provide decent performance. The best performing model is decided by the ROC which is 0.995 for the RF model, which is comparatively better than the rest of the models. The training and testing are shared as 80:20 basis on the available dataset. The best performing model is represented in Fig. 5.

The ten-fold validation of the proposed model is represented in Table III. The GB model outperforms the other models with the best of parameters in the ten-fold validation.

Table IV shows the Mean, SD and the confidence intervals (CI) of various parameters used in the machine learning models.

The training time and computational complexity of all the models is presented in Table V. FL and VAE shows high

TABLE II
CLASSIFICATION REPORT OF THE VARIOUS ML MODELS

Method	Accuracy	Precision	Recall	F1-Score	ROC
LR	0.89	0.88	0.89	0.88	0.91
DT	0.99	0.99	0.99	0.99	0.99
RF	0.99	0.99	0.99	0.99	0.995
AB	0.99	0.99	0.99	0.99	0.987
GB	0.99	0.99	0.99	0.99	0.993

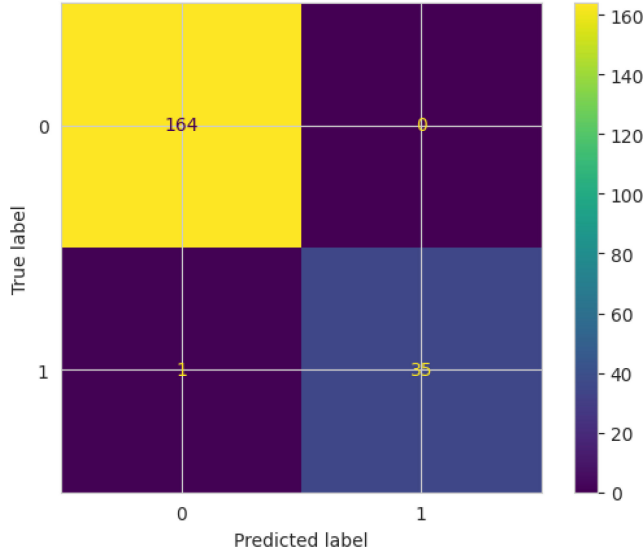


Fig. 5. Confusion Matrix for the best performing RF model.

TABLE III
PERFORMANCE COMPARISON OF THE VARIOUS ML MODELS IN TEN-FOLD

Model	Accuracy	Precision	Recall	F1-Score	ROC
LR	0.903	0.897	0.903	0.897	0.942
DT	0.992	0.992	0.992	0.992	0.979
RF	0.99	0.973	0.972	0.973	0.974
AB	0.997	0.997	0.997	0.997	0.991
GB	0.998	0.998	0.998	0.998	0.997

TABLE IV
DESCRIPTIVE STATISTICS FOR CLASSIFICATION METRICS ACROSS METHODS

Metric	Mean	SD	95% CI
Accuracy	0.970	0.0447	[0.9145, 1]
Precision	0.968	0.0492	[0.9069, 1]
Recall	0.970	0.0447	[0.9145, 1]
F1-score	0.968	0.0492	[0.9069, 1]
ROC	0.975	0.0365	[0.9297, 1]

computational complexity, whereas the LR shows the low computational complexity. It also indicates the complexity due to communication cost for the FL model in comparison with the other models, throwing a new challenge.

The FL model and auto encoders were also deployed in the proposed work, showed significant performance. The FL models exhibit an ROC of 0.999. The other parameters, such as accuracy is recorded as 0.975, with precision almost closer to 1. But the recall was 0.86 which was a major drawback. The auto encoders recorded accuracy and f1-score of 0.88 and AUC of 0.9217. The results are tabulated in Table VI. The FL provided AUC of 0.999 and showed the stability. The VAE is

TABLE V
COMPARISON OF TRAINING TIME AND COMPUTATIONAL COMPLEXITY

Model	Training Time	Computational Complexity
RF	Medium-High	$O(n \cdot m \cdot \log m)$ per tree
LR	Low	$O(n \cdot m)$
DT	Low-Medium	$O(n \cdot m \cdot \log m)$
AB	Medium	$O(T \cdot n \cdot m)$
GB	Medium-High	$O(T \cdot n \cdot m \cdot \log m)$
FL	High	$O(R \cdot n \cdot m) + \text{communication cost}$
VAE	High	$O(E \cdot n \cdot m \cdot h)$

TABLE VI
CLASSIFICATION REPORT OF FL AND AUTO ENCODERS

Method	Accuracy	Precision	Recall	F1-Score	ROC
FL	0.975	0.999	0.864	0.9275	0.999
VAE	0.88	0.88	0.88	0.88	0.9217

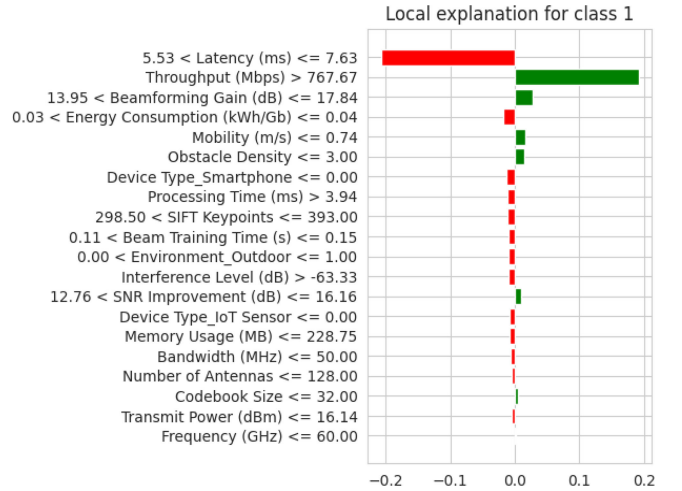


Fig. 6. LIME pyplot for the feature polarization.

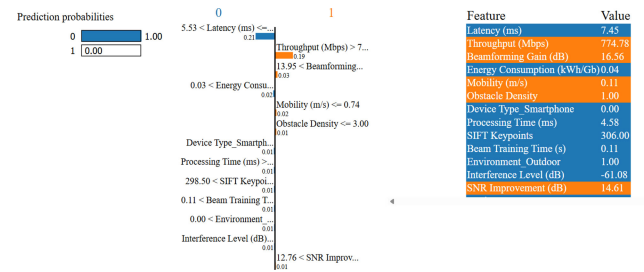


Fig. 7. LIME notebook for detailed feature analysis.

set up for the 50 epochs with batch size of 32, with learning rate of 0.001. The encoder is activated by “rectified linear unit (RELU)” and the decoder is activated by “Sigmoid” function. The model is optimized by the “Adam” optimizer and the loss estimation is computed for the “Mean Squared Error.”

LIME pyplot depicts feature polarity, while the notebook presents the corresponding weights. As a local surrogate model, LIME explains a single instance using lasso-based approximation. Green indicates positive and red shows negative impact. Fig. 6 shows throughput as the most positive feature and latency as the key negative feature to minimize. Fig. 7 presents feature weights, with latency having the highest impact (0.21).

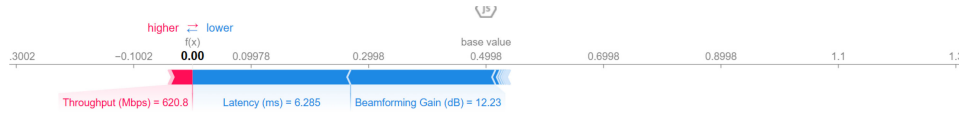


Fig. 8. SHAP force plot feature evaluation.

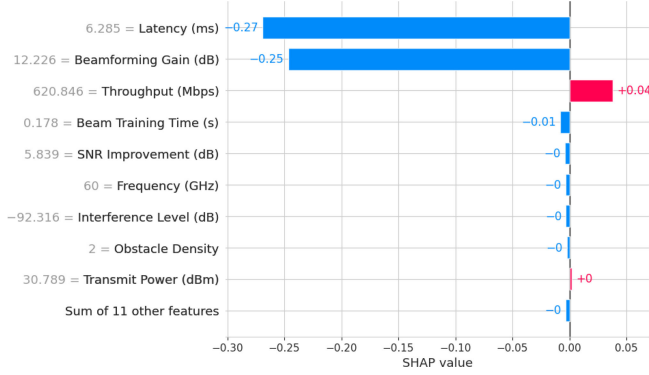


Fig. 9. SHAP box plot feature evaluation.

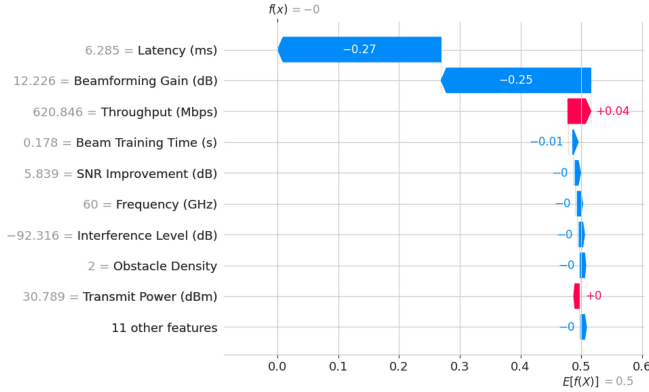


Fig. 10. SHAP waterfall plot feature evaluation.

The SHAP represents the global surrogacy. The force plot, which represents positive and negative features. Features which are represented closer to the boundary are the most important features. According to Fig. 8, it is proven that latency and throughput are the most important features that are represented at the boundary of the force plot.

The box plot which shows the features, weights and the corresponding polarity. The latency with a weight of -0.27 , with negative polarity, is the most important feature. The throughput which has the weight of 0.04 has positive weight of importance. The box plot is represented in Fig. 9. This diagram shows the latency is lower the better and should be kept minimum and throughput should be maximum the better.

Fig. 10 represents a waterfall plot, which ensures the order of importance of the features, polarity, and weights. The most important features that affect the target attribute optimization, are latency and throughput.

The final decision plot represented in Fig. 11, ensures the influence of latency in the target prediction. It shows the major change in the decision toward low optimization, when latency reaches more that reduces the performance. Since latency is

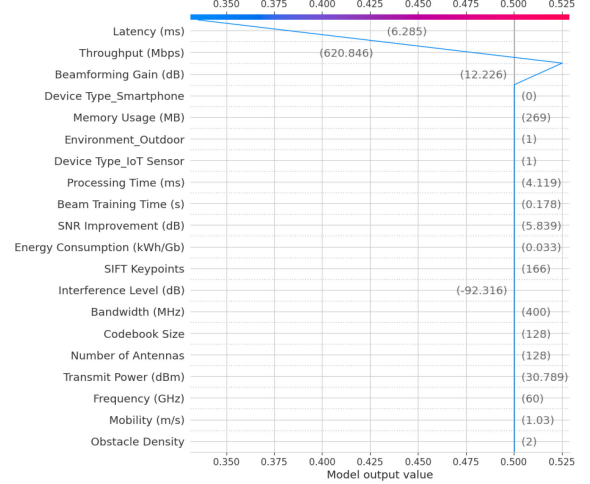


Fig. 11. SHAP decision plot showing cumulative feature influence.

a negative feature, it should be lower. The throughput is a positive feature, required to be maximum. Thus, the SHAP diagrams do the global analysis of features and provide the insight of the two important features latency and throughput, and their impact on determining the target optimization of IoT enabled 6G infrastructure. However, while the use of SHAP and LIME significantly improves transparency and interpretability, these explanations may expose feature sensitivities, highlighting a tradeoff between interpretability and privacy that must be considered when deploying the system in real-world 6G-IoT settings.

Federated models work with raw data on local devices, reducing privacy risks but still enabling joint model training. FL balances between privacy and accuracy using techniques, such as differential privacy, giving organizations flexibility to meet both performance and compliance goals.

The primary objectives of the research are optimization through minimization and maximization of parameters, such as latency, to be minimized and the throughput which needs to be maximized. The quantum of minimization and maximization is illustrated by XAI. FL enhances the performance, privacy and stability of the proposed work.

VI. CONCLUSION

The proposed work analyzes the impact of the features in optimizing intelligent 6G networks assisted by IoT. The results of XAI show that the optimization could be achieved by diminishing the latency and enhancing the throughput. The ML models perform better with an accuracy of 0.99 and an ROC of 0.995 . The autoencoder model measured with accuracy levels of 0.88 . FL recorded the best ROC of 0.999 . Thus, the framework combines the efforts of ML, FL and autoencoders,

which makes the proposed work suitable for the optimization of any such networks, such as SD-WAN, with the presence of FL, XAI and VAE.

REFERENCES

- [1] S. Selvarajan, H. Manoharan, A. O. Khadidos, A. O. Khadidos, A. M. Alshareef, and A. Alsobhi, "Secured 6G communication for consumer electronics with advanced artificial intelligence algorithms," *IEEE Trans. Consum. Electron.*, vol. 70, no. 3, pp. 5711–5718, Aug. 2024.
- [2] M. H. Alsharif, A. H. Kelechi, M. A. Albreem, S. A. Chaudhry, M. S. Zia, and S. Kim, "Sixth generation (6G) wireless networks: Vision, research activities, challenges and potential solutions," *Symmetry*, vol. 12, no. 4, p. 676, 2020.
- [3] M. Al-Ali and E. Yaacoub, "Resource allocation scheme for eMBB and uRLLC coexistence in 6G networks," *Wireless Netw.*, vol. 29, no. 6, pp. 2519–2538, 2023.
- [4] M. Das et al., "Synergy of 6G technology and IoT networks for transformative applications," *Int. J. Commun. Syst.*, vol. 37, no. 14, 2024, Art. no. e5869.
- [5] A. Jahid, M. H. Alsharif, and T. J. Hall, "The convergence of blockchain, IoT and 6G: Potential, opportunities, challenges and research roadmap," *J. Netw. Comput. Appl.*, vol. 217, Aug. 2023, Art. no. 103677.
- [6] X. Zhou, W. Liang, J. She, Z. Yan, I. Kevin, and K. Wang, "Two-layer federated learning with heterogeneous model aggregation for 6g supported Internet of Vehicles," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 5308–5317, Jun. 2021.
- [7] Y. Lu, X. Huang, K. Zhang, S. Maharjan, and Y. Zhang, "Low-latency federated learning and blockchain for edge association in digital twin empowered 6G networks," *IEEE Trans. Ind. Informat.*, vol. 17, no. 7, pp. 5098–5107, Jul. 2021.
- [8] S. Roy, H. Chergui, and C. Verikoukis, "Explanation-guided fair federated learning for transparent 6G RAN slicing," *IEEE Trans. Cogn. Commun. Netw.*, vol. 10, no. 6, pp. 2269–2281, Dec. 2024.
- [9] W. Ziegler, "A landscape analysis of standardisation in the field of artificial intelligence," *J. ICT Stand.*, vol. 8, no. 2, pp. 151–184, 2020.
- [10] N. Kaur and L. Gupta, "Securing the 6G–IoT environment: A framework for enhancing transparency in artificial intelligence decision-making through explainable artificial intelligence," *Sensors*, vol. 25, no. 3, p. 854, 2025.
- [11] M. W. P. Maduranga, V. Tilwari, R. Rathnayake, and C. Sandamini, "AI-enabled 6G Internet of Things: Opportunities, key technologies, challenges, and future directions," *Telecom*, vol. 5, no. 3, pp. 804–822, 2024.
- [12] A. Kovari, "Synergizing 6G networks, IoT, and AI: Paving the way for next-generation intelligent ecosystems," *J. Eng. Sci. Technol.*, vol. 20, no. 1, pp. 114–128, 2025.
- [13] A. Blika et al., "Federated learning for enhanced cybersecurity and trustworthiness in 5G and 6G networks: A comprehensive survey," *IEEE Open J. Commun. Soc.*, vol. 6, pp. 3094–3130, 2025.
- [14] H. Sun et al., "Advancing 6G: Survey for explainable AI on communications and network slicing," *IEEE Open J. Commun. Soc.*, vol. 6, pp. 1372–1412, 2025.
- [15] M. Arana-Catania et al., "Explainable reinforcement and causal learning for improving trust to 6G stakeholders," *IEEE Open J. Commun. Soc.*, vol. 6, pp. 4101–4125, 2025.
- [16] J. Parra-Ullauri et al., "Lifecycle management of trustworthy ai models in 6g networks: The reason approach," *IEEE Wireless Commun.*, vol. 32, no. 2, pp. 42–51, Apr. 2025.
- [17] D. C. Nguyen et al., "Federated learning for Industrial Internet of Things in future industries," *IEEE Wireless Commun.*, vol. 28, no. 6, pp. 192–199, Dec. 2021.
- [18] Z. Chen et al., "A trustworthy decentralized federated learning framework for consumer electronics: Mitigating large-scale AIoT heterogeneity through transfer knowledge integration," *IEEE Trans. Consum. Electron.*, vol. 71, no. 1, pp. 1918–1925, Feb. 2025.
- [19] Q. Sun, J. Zeng, L. Dai, Y. Hu, and L. Tian, "XAI-based framework for protocol anomaly classification and identification to 6G NTN with drones," *Drones*, vol. 9, no. 5, p. 324, 2025.
- [20] A. A. S. Mohammad et al., "Strategies for applying interpretable and explainable AI in real world IoT applications," *Discov. Internet Things*, vol. 5, no. 1, pp. 1–17, 2025.
- [21] T. B. Donmez, M. Kutlu, M. Mansour, and M. Z. Yildiz, "Explainable AI in action: A comparative analysis of hypertension risk factors using SHAP and LIME," *Neural Comput. Appl.*, vol. 37, no. 5, pp. 4053–4074, 2025.
- [22] A. M. Al Shahrani, "IoT enabled intelligent receiver synchronization system for long-range 6G consumer technology," *IEEE Trans. Consum. Electron.*, vol. 70, no. 3, pp. 5747–5755, Aug. 2024.
- [23] J. Wu, J. Zhang, M. Bilal, F. Han, N. Victor, and X. Xu, "A federated deep learning framework for privacy-preserving consumer electronics recommendations," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2628–2638, Feb. 2024.
- [24] X. Wang, J. Lyu, J. D. Peter, and B.-G. Kim, "Privacy-preserving AI framework for 6G-enabled consumer electronics," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 3940–3950, Feb. 2024.
- [25] D. Das, P. Chatterjee, S. Banerjee, U. Ghosh, and M. S. Al-Numay, "Blockchain-enabled federated learning for security and privacy in consumer electronics devices," *IEEE Trans. Consum. Electron.*, vol. 71, no. 1, pp. 2262–2270, Feb. 2025.
- [26] S. Maitra, V. Mishra, P. Verma, M. Chopra, and P. Nath, "Sampling-variational auto encoder-ensemble: In the quest of explainable artificial intelligence," in *Proc. Int. Conf. Elect. Inf. Technol. (IEIT)*, 2023, pp. 43–50.
- [27] S. Alsubai, M. Umer, N. Innab, S. Shiaeles, and M. Nappi, "Multi-scale convolutional auto encoder for anomaly detection in 6G environment," *Comput. Ind. Eng.*, vol. 194, Aug. 2024, Art. no. 110396.
- [28] X. Xiang, H. Yu, Y. Wang, and G. Wang, "Stable local interpretable model-agnostic explanations based on a variational autoencoder," *Appl. Intell.*, vol. 53, no. 23, pp. 28226–28240, 2023.
- [29] N. Umashankar, K. Sai Geethanjali, I. Rajesh, M. Bharathi, and S. Karthik, "Advancements in AI and ML for enhanced security and performance in 6G networks," in *6G Cyber Security Resilience: Trends and Challenges*. Cham, Switzerland: Springer, 2025, pp. 99–120.
- [30] "6G IoT intelligent management dataset." Ziya07. 2023. [Online]. Available: <https://www.kaggle.com/datasets/ziya07/6g-iot-intelligent-management-dataset>